

Tracescope

Tracescope is a group project created to improve the PCB (Printed Circuit Board) design workflow. The app's story begins in the Inspirit AI summer camp. The camp focused on creating apps that help the community using AI. My team and I created the project in Lovable.

Tracescope is an app focused on PCB checking. The app uses AI to inspect Gerber files and note things that most DRCs wouldn't spot. The user will enter a description of their PCB, upload their files, and hit analyze. Within seconds, the app provides in-depth explanations of potential failure points. A problem is given a severity of warning or severe. The app also provides an overlay with arrows pointing in the general direction of the problem. If the user dislikes the placement, they can edit the arrows themselves by adding more or removing arrows. The arrows can also be realigned to ensure that the user can come back and understand where they may have made a mistake.

If the user ever wants to come back to a previously inspected board, the app includes a history tab with all inspections listed. The user can revisit the board and view its problems again. The user can also chat with a built-in AI chatbot that explains whatever inquiries the user may have.

The built-in AI chatbot has strict rules on appropriate use. If the user asks a question that is not related to PCBs, the AI will redirect the user. To use it, enter the name of the chat and ask a question. The chatbot also has a section with all old chats.

There is also a community page. Engineers can post their designs, and the inspector will suggest that the user look at relevant designs created by other users. The users are warned on the front page that people can see what they post and that their designs will be screened for inappropriate content. There is also a Gerber viewer that lets users view their boards.

There are also accounts and account settings in the app. Just sign in, and you will be prompted with a username and theme settings menu. To sign in, you can use email and password with a two-step authentication process. Or the user can use Apple or Google to sign in as well. It is taken seriously that the user's account information is secure. All information is stored in the app's backend on Lovable's database.

The link to begin using the app is: [TraceScope — AI-powered PCB inspection](#)